Complex Document for RAG Parsing Tests

Synthetic 15-page PDF with paragraphs, simple and complex tables, diagrams, scanned-form style image, metadata examples, and production RAG edge cases.

Story Line

Three client teams - Arka Finance, BlueLeaf Retail, and CityRide Mobility - are migrating contracts, policies, support records, and operational reports into a single RAG platform. Each team has different document types, access rules, and parsing challenges. The RAG system must answer questions with citations while ensuring that one client never sees another clients data.

This PDF is intentionally designed to test document loaders, PDF parsers, OCR workflows, table extraction, chunking strategies, metadata preservation, and source citation quality.

Key statement: RAG does not train the model. RAG gives the model the right context before answering.

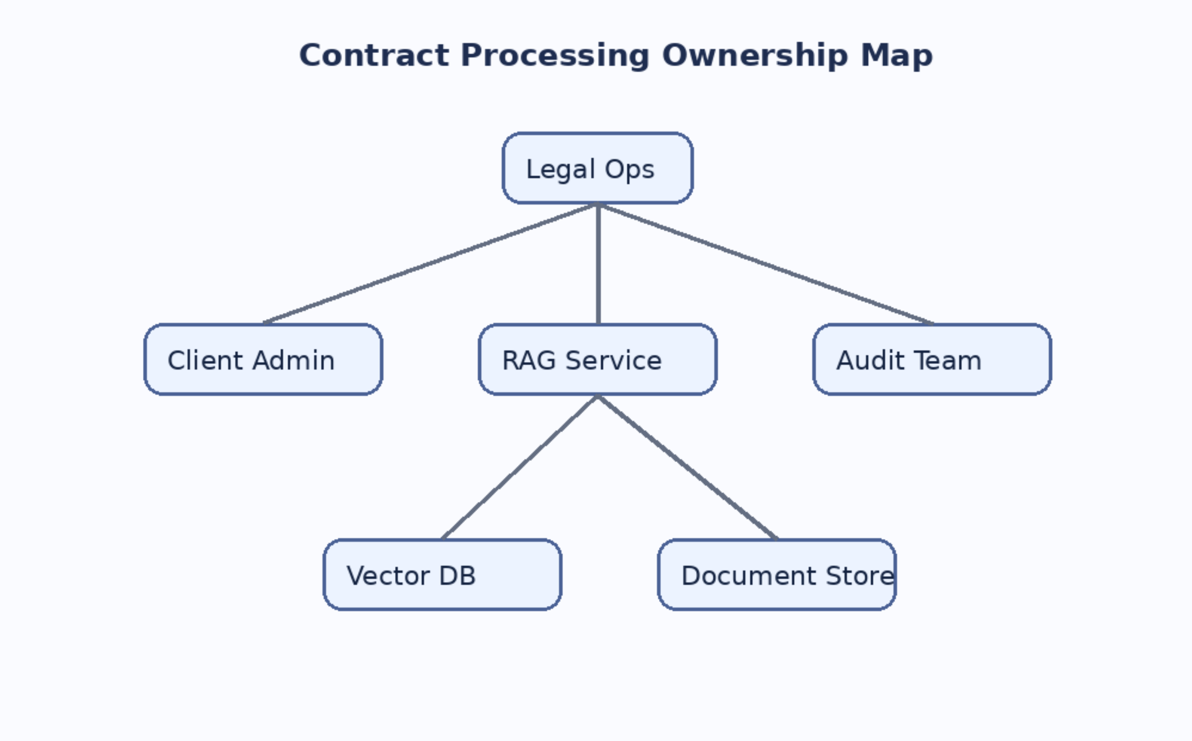
Section	Parsing challenge	Why it matters for RAG
Contracts	Dense legal text, clause numbers, cross references	Need section-aware chunks and citations
Tables	Merged headers, numeric columns, footnotes	Need row/column preservation
Images	Architecture diagram, heatmap, scanned form	Need OCR or multimodal extraction
Multi-tenant metadata	client_id, document_id, contract_group_id	Need access-safe retrieval

1. Business Scenario

The platform team is building a document intelligence layer for contract operations. Legal users need answers such as: Which contracts renew automatically? Which agreements have a 60-day termination notice? Which documents mention data residency in India? These questions require retrieval across multiple document types and sometimes across multiple related documents of a single client.

Support teams also want answers from SOPs, markdown runbooks, HTML exports, chat transcripts, and CSV logs. A single parser is not enough because the corpus contains born-digital PDFs, scanned forms, HTML, markdown, and spreadsheet-like tables.

Client	Main document family	Risk	Primary users	Typical question
Arka Finance	MSA, audit addendum, DPA	High	Legal and compliance	What audit rights exist in the DPA?
BlueLeaf Retail	MSA, pricing, support SLA	Medium	Procurement	Which clause controls renewal pricing?
CityRide Mobility	Ops reports, incident logs, SOPs	Medium	Operations	Which depot had repeated battery incidents?



Caption: Ownership map showing how client admins, legal teams, audit teams, document storage, vector storage, and the RAG service interact.

2. Raw Data Inventory

The ingestion team receives files from multiple sources. Some files are clean and digital; others contain scanned pages, rotated tables, missing metadata, embedded images, or formula-generated values. The inventory below intentionally includes different formats and extraction expectations.

Asset ID	Client	Format	Pages/Rows	Expected extraction	Notes
ARK-MSA-001	Arka Finance	PDF	32 pages	Clauses, tables, signatures	Born-digital text with footer references
ARK-DPA-002	Arka Finance	DOCX	18 pages	Headings, lists, tables	Track changes removed before ingestion
BLR-PRICE-003	BlueLeaf Retail	XLSX	12 sheets	Rate cards, discounts	Merged cells and formulas
BLR-SLA-004	BlueLeaf Retail	PDF	9 pages	SLA table, escalation matrix	Table crosses page boundary
CRM-OPS-005	CityRide Mobility	CSV	48,200 rows	Incident records	Needs column type normalization
CRM-FORM-006	CityRide Mobility	Scanned PDF	4 pages	OCR fields, checkbox	Low contrast and slight rotation
KB-RUN-007	Shared	Markdown	14 files	Runbook sections	Good for heading-based chunks
WEB-FAQ-008	Shared	HTML	23 pages	FAQ content, links	Needs boilerplate removal

Parsing difficulty increases when a document combines multiple layouts: paragraph text, tables, images, and notes. Good parsers preserve structure; weak parsers flatten everything into a noisy text stream.

3. Contract Excerpt: Dense Legal Text

Clause 7.2 - Confidential Information. The Receiving Party shall protect Confidential Information using at least the same degree of care that it uses to protect its own confidential materials, but in no event less than reasonable care. Confidential Information includes technical documents, pricing schedules, user lists, API keys, operational procedures, security reports, incident response notes, product roadmaps, and any derived analysis prepared by the Receiving Party.

Clause 7.3 - Exclusions. Confidential Information does not include information that is publicly available without breach, independently developed without reference to the Disclosing Party material, received lawfully from a third party, or approved for release in writing. The burden of proving an exclusion remains with the Receiving Party.

Clause 9.1 - Data Residency. If the Statement of Work identifies a specific processing region, the service provider shall process production records only within that region unless the client approves a transfer in writing. Disaster recovery replicas may be stored in a secondary region if encryption at rest and access logging are enabled.

Clause 11.4 - Termination Assistance. Upon termination, the provider shall make client data available for export for a period of thirty days. After the export period, the provider may delete production data according to the deletion schedule unless a legal hold applies.

Clause	Short name	Owner	Trigger	Required action	Citation hint
7.2	Confidentiality	Receiving Party	Receipt of confidential material	Protect information with reasonable care and equivalent internal controls	Page 4, clause 7.2
9.1	Data residency	Service Provider	SOW specifies region	Process production records only in approved region	Page 4, clause 9.1
11.4	Termination assistance	Service Provider	Agreement termination	Provide export window for 30 days	Page 4, clause 11.4

Parser note: clause numbering is important. Chunking should not remove or split clause identifiers because users often ask questions using clause numbers.

4. Simple Table: Policy Rules

The following table is intentionally simple. It should be correctly extracted by most PDF table parsers. It tests basic row and column detection, short text values, and numeric values.

Policy Area	Rule	Owner	Review Cycle
Refunds	Refund requests must be raised within 7 days of purchase.	Customer Support	Quarterly
Data Export	Client admins can request export in CSV or JSON format.	Platform Ops	Monthly
Access Review	Privileged access must be reviewed every 30 days.	Security	Monthly
Incident Response	Critical incidents require acknowledgement within 30 minutes.	SRE	After each incident
Vendor Review	High-risk vendors require annual risk assessment.	Procurement	Annual

Narrative Around the Table

This table appears between paragraphs, which is common in enterprise documents. A good parser should preserve the paragraph before the table, the table content, and the paragraph after the table in the correct reading order.

For RAG systems, simple tables are often converted into row-wise text chunks such as: Policy Area: Refunds; Rule: Refund requests must be raised within 7 days; Owner: Customer Support.

5. Complex Table: SLA, Escalation, and Credits

This table uses long cells and operational footnotes. It is designed to test whether the parser can retain header hierarchy and map values to the correct columns.

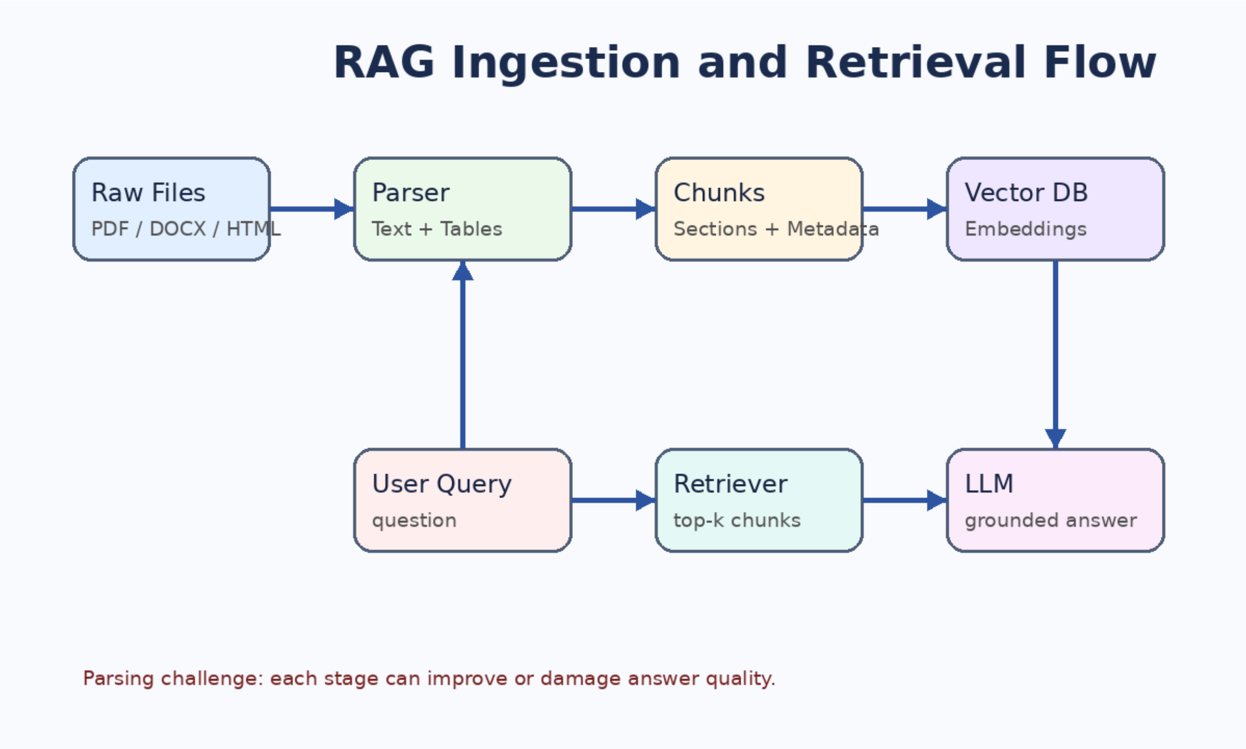
Service Category	Severity	Response Target	Resolution Target	Service Credit	Evidence Required
API Availability	P1 - complete outage	15 minutes	4 hours	10 percent monthly fee credit	Monitoring logs and incident ticket
API Availability	P2 - degraded performance	30 minutes	8 hours	5 percent monthly fee credit	Latency dashboard and affected endpoint list
Data Pipeline	P1 - ingestion stopped	20 minutes	6 hours	8 percent pipeline fee credit	Queue depth, failed job IDs, replay report
Data Pipeline	P3 - delayed batch	4 hours	Next business day	No automatic credit	Batch ID and SLA exception note
Support	P2 - urgent support	1 hour	1 business day	2 percent support fee credit	Support ticket with timestamps

Footnote A: Service credits are not cumulative across the same service category for the same calendar month. Footnote B: Credits do not apply when delay is caused by client-side network restrictions, missing credentials, or force majeure events.

Parser note: table footnotes should remain attached to the table. If the footnote is separated, an answer about credits may become incomplete or misleading.

6. Image: RAG Architecture Diagram

The diagram below represents the ingestion and retrieval flow. Some PDF parsers ignore images completely, while others extract image metadata but not the text inside the image. For production use, image content may require OCR or multimodal extraction.



Caption: A simplified RAG pipeline showing raw files, parsing, chunking, embeddings, vector database, retriever, LLM, and grounded answer generation.

Image element	Potential extraction method	Expected output
Box labels	OCR or vision model	Raw Files, Parser, Chunks, Vector DB
Arrows	Layout or vision understanding	Processing order
Caption	Normal PDF text extraction	Description of the diagram

7. Markdown Runbook Excerpt

Markdown files are often easier to parse because headings and code blocks are explicit. However, when markdown is exported to PDF, the structure may become visual rather than semantic.

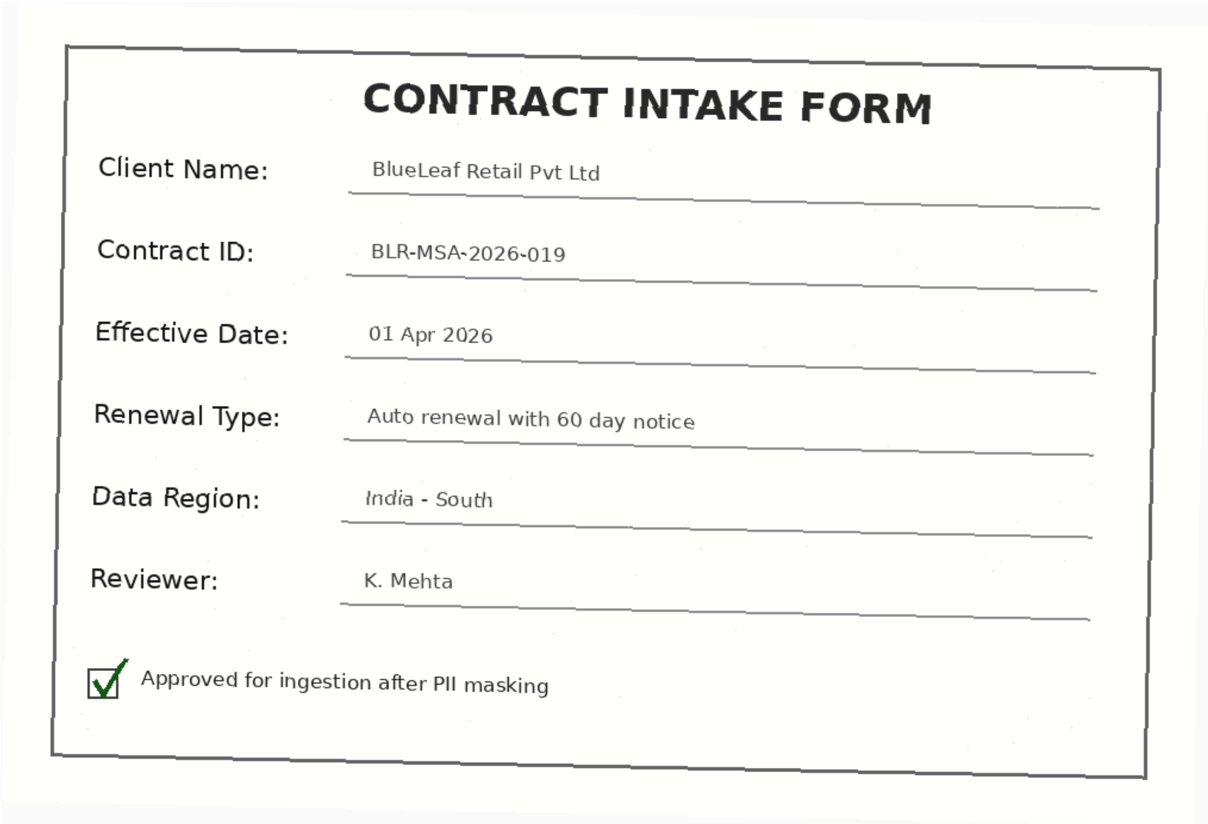
```
# Runbook: Contract Ingestion Failure

## Symptoms
- Upload status remains `processing` for more than 30 minutes.
- Worker logs show repeated timeout errors.
- Vector count does not increase for the affected document.

## Recovery Steps
1. Confirm the object exists in document storage.
2. Re-run parser with `safe_mode=True`.
3. Rebuild chunks with the previous chunking configuration.
4. Compare chunk count with the last successful ingestion.
5. Trigger re-embedding only for changed chunks.
```

A good parser should preserve code block boundaries and avoid mixing numbered steps with surrounding prose. In RAG, runbooks are useful because users often ask operational questions that require exact steps.

Runbook field	Value	Parsing expectation
Title	Contract Ingestion Failure	Should become section title metadata
Symptoms	Five bullet items	Should remain as list or separate lines
Recovery Steps	Numbered sequence	Order must be preserved

8. Scanned Form and OCR Challenge

This page contains a scanned-form style image. Text inside the form is not normal selectable PDF text. A simple text parser may miss it entirely. OCR-based systems should extract labels, values, and checkboxes from the image.



CONTRACT INTAKE FORM

Client Name: BlueLeaf Retail Pvt Ltd

Contract ID: BLR-MSA-2026-019

Effective Date: 01 Apr 2026

Renewal Type: Auto renewal with 60 day notice

Data Region: India - South

Reviewer: K. Mehta

☑ Approved for ingestion after PII masking

Caption: The form includes fields such as Client Name, Contract ID, Effective Date, Renewal Type, Data Region, Reviewer, and a checkbox approval statement.

Field	Expected OCR value	Validation rule
Client Name	BlueLeaf Retail Pvt Ltd	Must match known client list
Contract ID	BLR-MSA-2026-019	Pattern: client-code + type + year + sequence
Effective Date	01 Apr 2026	Date parser should normalize to ISO format
Approval checkbox	Checked	Boolean extraction required

9. Multiple Documents for One Client

BlueLeaf Retail has five related contract documents. If these documents are indexed independently without relationship metadata, the retriever may miss cross-document context. The platform uses a contract_group_id to connect related documents.

document_id	document_type	contract_group_id	effective_date	relationship
BLR-MSA-001	Master Service Agreement	BLR-ACME-2026	2026-04-01	Base commercial agreement
BLR-NDA-002	NDA	BLR-ACME-2026	2026-04-01	Confidentiality terms
BLR-SOW-003	Statement of Work	BLR-ACME-2026	2026-04-15	Project scope and deliverables
BLR-PRICE-004	Pricing Amendment	BLR-ACME-2026	2026-05-01	Updated pricing and discount tiers
BLR-SLA-005	Support SLA	BLR-ACME-2026	2026-05-10	Support response and credits

Example question: What does the BlueLeaf agreement say about termination assistance and pricing changes? A good retriever may need chunks from the MSA and the Pricing Amendment together.

Relationship metadata allows retrieval across related documents without mixing unrelated client data.

10. Multi-tenant Retrieval and Access Control

In a multi-tenant RAG system, each document, chunk, embedding, and retrieval request should be associated with a tenant identifier. Client isolation should happen before the LLM receives any context.

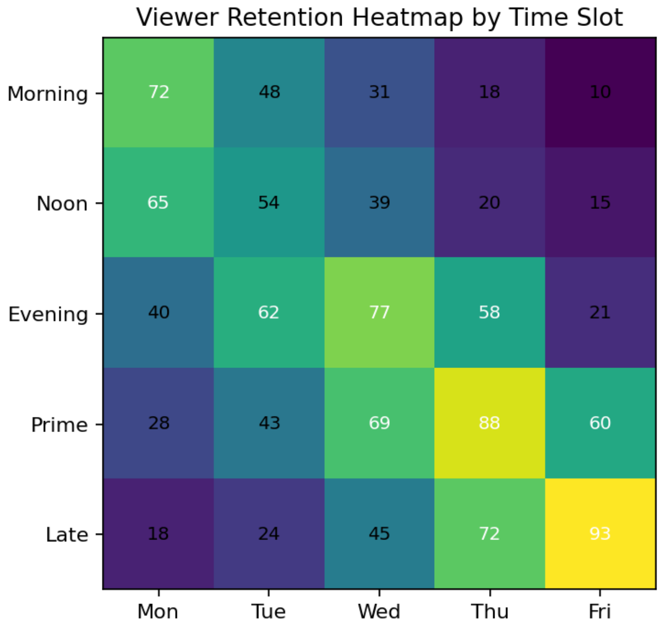
Layer	Control	Failure mode if missing
Authentication	Identify user and organization	Unknown user may access application
Authorization	Map user to client_id and role	User may retrieve another clients documents
Vector retrieval	Filter by client_id or namespace	Retriever may return unauthorized chunks
Prompt construction	Send only allowed context	LLM may see sensitive data
Audit logging	Record query, source chunks, user_id	No traceability during incident review

Correct flow: authenticate user, get client_id from backend, retrieve only matching chunks, build prompt with authorized context, return answer with citations.

Wrong flow: retrieve from all clients and tell the model to ignore unauthorized content. This is unsafe because the LLM should not be used as the access-control boundary.

11. Analytics Image: Retention Heatmap

The heatmap below simulates viewership retention by time slot. A parser that ignores chart images will miss useful business context. OCR can extract axis labels and numbers, but chart understanding may require a vision model.



Observation	Supporting value	Scheduling implication
Prime Friday performs best	93 retention score	Place premium content in Friday late/prime slots
Early week late slots underperform	10 to 21 score range	Avoid launching new series in weak slots
Evening improves midweek	77 on Wednesday evening	Use midweek evening for discovery content

For RAG, chart captions should be indexed. If chart values are critical, extract them into structured metadata or a companion table.

12. Evaluation Dataset

A RAG system should be evaluated separately for retrieval quality and answer quality. This page includes a miniature evaluation set with expected source references. It is useful for testing whether citations point to the correct section.

Test ID	Question	Expected source	Expected answer element	Failure signal
Q-001	What is the refund window?	Policy Rules table	7 days	Answer says 30 days
Q-002	Which clause covers data residency?	Clause 9.1	Approved processing region	No clause citation
Q-003	What is P1 API response target?	SLA table	15 minutes	Wrong severity row
Q-004	Which documents belong to BLR-ACME-2026?	Relationship table	Five related documents	Only one document returned
Q-005	Who owns access review?	Policy Rules table	Security	Owner missing

Evaluation should include adversarial questions, unrelated questions, and questions that require multi-hop retrieval across related documents.

13. Edge Cases for Parsers

The following cases often break real document ingestion pipelines. They are included here as guidance for testing parser quality before moving to embeddings and vector storage.

Edge case	Example	Recommended handling
Repeated headers and footers	Page number, confidentiality banner	Remove or store separately as metadata
Hyphenated line breaks	termi- nation assistance	Normalize during cleaning
Rotated tables	Landscape appendix in PDF	Use layout-aware parser or OCR
Merged cells	Pricing table with grouped plans	Preserve hierarchy in row text
Scanned signatures	Signature block as image	OCR if text is needed; store image reference
Boilerplate navigation	Website header and footer	Use boilerplate removal
Duplicate chunks	Same policy in FAQ and PDF	Deduplicate using source and hash

Bad parsing creates bad chunks. Bad chunks create bad retrieval. Bad retrieval creates bad answers.

15. Final Ingestion Checklist

Use this checklist before sending parsed content into chunking and embeddings. It helps identify whether the data is ready for production RAG.

Checklist item	Status to verify	Why it matters
Text extraction	Paragraphs are readable and ordered	Prevents noisy chunks
Table extraction	Rows, columns, headers, and footnotes preserved	Protects factual answers
Image handling	Captions indexed and OCR done if required	Avoids missing visual information
Metadata	source, page, client_id, document_id captured	Enables citations and access control
Chunking	Chunks preserve meaning and section boundaries	Improves retrieval relevance
Access control	Retrieval filters use authenticated client_id	Prevents data leakage
Evaluation	Golden questions tested with citations	Measures real answer quality

Summary: RAG is about knowledge access. Fine-tuning is about behavior adaptation. For document intelligence, parsing quality is the foundation. If extraction is weak, no embedding model or LLM can fully fix the missing context.

End of synthetic 15-page parsing test document.

Appendix A: Complex Clause Responsibility Matrix

Grouped clauses with owner/backup split inside the same responsibility cell. This page is useful for testing row grouping, merged-looking labels, split responsibility cells, and long evidence text.

Parsing challenge: preserve row boundaries, nested headers, split cells, grouped labels, numeric values, and footnotes/context around the table.

and a split responsibility cell where owner and backup are shown inside the same row.

Clause Group	Obligation	Responsible Team	Trigger	Evidence Required	Risk
Data Protection	Delete client data after contract termination unless retention is legally required.	**Owner** Compliance **Backup** Legal	Termination notice received	Deletion certificate + audit log export	High
	Notify client about any confirmed data incident within 72 hours.	**Owner** Security **Backup** DPO	Incident classified as confirmed breach	Incident report, timeline, notification proof	Critical
Billing	Apply annual platform fee adjustment only after renewal confirmation.	**Owner** Finance **Backup** CSM	Renewal order approved	Approved renewal sheet + invoice draft	Medium
	Do not bill inactive campuses during suspension period.	**Owner** Revenue Ops **Backup** Finance	Campus status = suspended	ERP campus status export	High
Support	Provide P1 response within 30 minutes during school operating hours.	**Owner** Support L2 **Backup** Ops Manager	Ticket priority = P1	Ticket timestamps + agent assignment log	High
	Escalate unresolved P2 tickets after 4 business hours.	**Owner** Support L1 **Backup** Support L2	Ticket age > 4 business hours	Escalation log	Medium

Table 1: Added as complex parsing appendix for table extraction, OCR fallback, and layout-aware RAG testing.

Appendix B: Regional Pricing and Usage Add-on Matrix

Pricing table with multi-level headers, regional columns, add-on columns, billing rules, exception rows, and mixed numeric/text values.

Parsing challenge: preserve row boundaries, nested headers, split cells, grouped labels, numeric values, and footnotes/context around the table.

Plan	Student Volume	Annual Platform Charges		Usage Add-ons		Billing Rule
		India Region	International	SMS	WhatsApp	
Starter	0 - 2,000 students	**Base** INR 4.5L **Support** INR 60K	USD 7,200	INR 0.18/message	INR 0.42/message	Quarterly advance
Growth	2,001 - 10,000 students	**Base** INR 11L **Support** INR 1.4L	USD 18,000	INR 0.15/message	INR 0.38/message	50% advance + monthly usage
Enterprise	10,001+ students	**Base** Custom **Support** Included	Custom	Negotiated	Negotiated	Signed order form required
Exception	Government schools	Discount may apply after approval		No discount on pass-through cost		Requires CFO approval

Table 2: Added as complex parsing appendix for table extraction, OCR fallback, and layout-aware RAG testing.

Appendix C: Invoice Line Items with Tax Split

Invoice-style line item table with item groups, quantity, rate, CGST/SGST split, totals, and summary row. Useful for invoice parsing and tax extraction tests.

Parsing challenge: preserve row boundaries, nested headers, split cells, grouped labels, numeric values, and footnotes/context around the table.

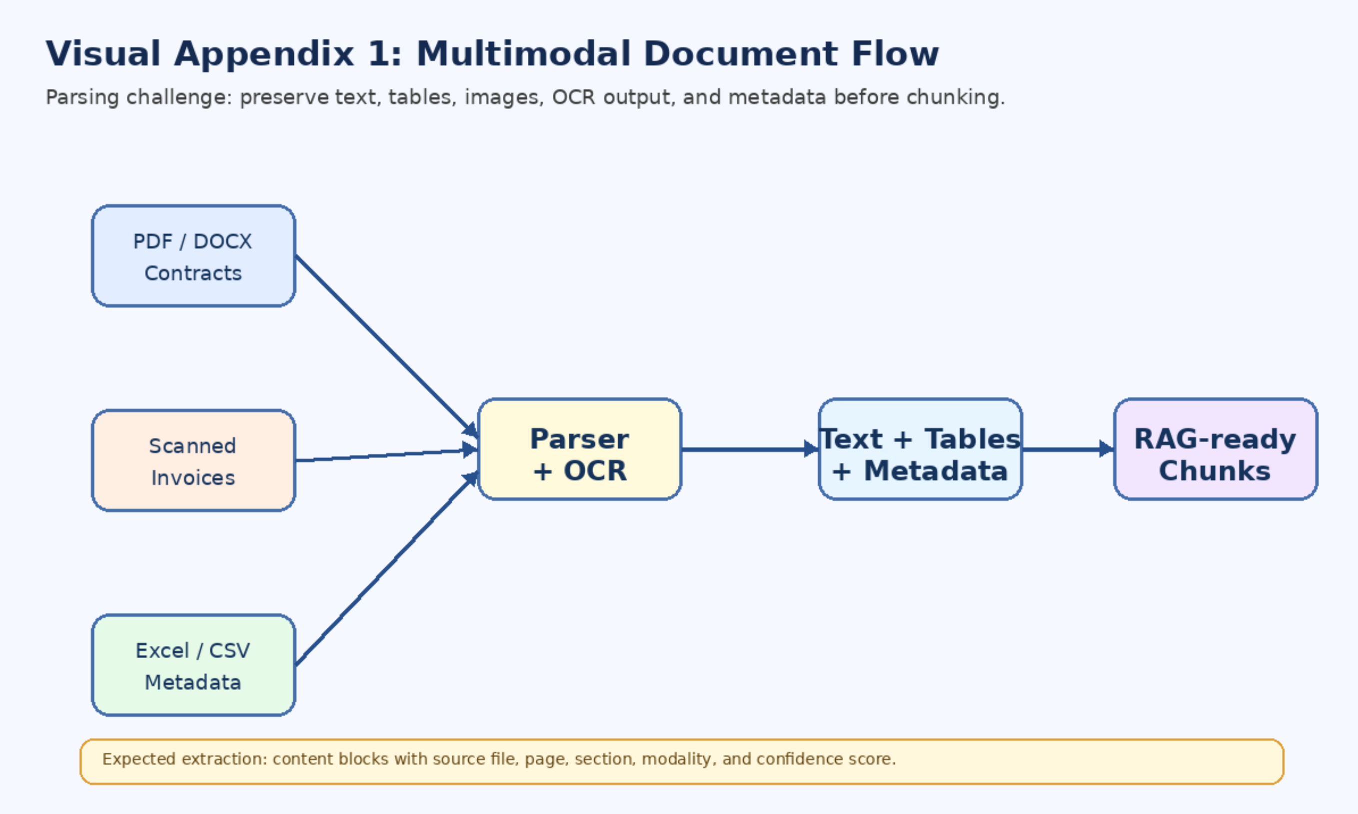
Item Group	Line Item	Qty	Rate	Tax Split		Total
				CGST	SGST	
ERP Platform	Annual School360 Enterprise Subscription - 12 campuses	1	INR 11,00,000	9%	9%	INR 12,98,000
	Parent communication add-on - estimated message pack	2,00,000 msgs	INR 0.38/msg	9%	9%	INR 89,680
Implementation	Data migration + training + go-live support	1	INR 2,40,000	9%	9%	INR 2,83,200
Summary	Subtotal and taxes	-	INR 14,16,000	INR 1,27,440	INR 1,27,440	INR 16,70,880

Table 3: Added as complex parsing appendix for table extraction, OCR fallback, and layout-aware RAG testing.

Appendix D: Multimodal Document Processing Flow

Diagram-style image showing how PDFs, DOCX files, scanned invoices, Excel/CSV metadata, parser/OCR, extracted text/tables/metadata, and RAG-ready chunks connect.

Parsing challenge: image text, arrows, labels, and captions may not appear in normal PDF text extraction. OCR or multimodal parsing may be required.

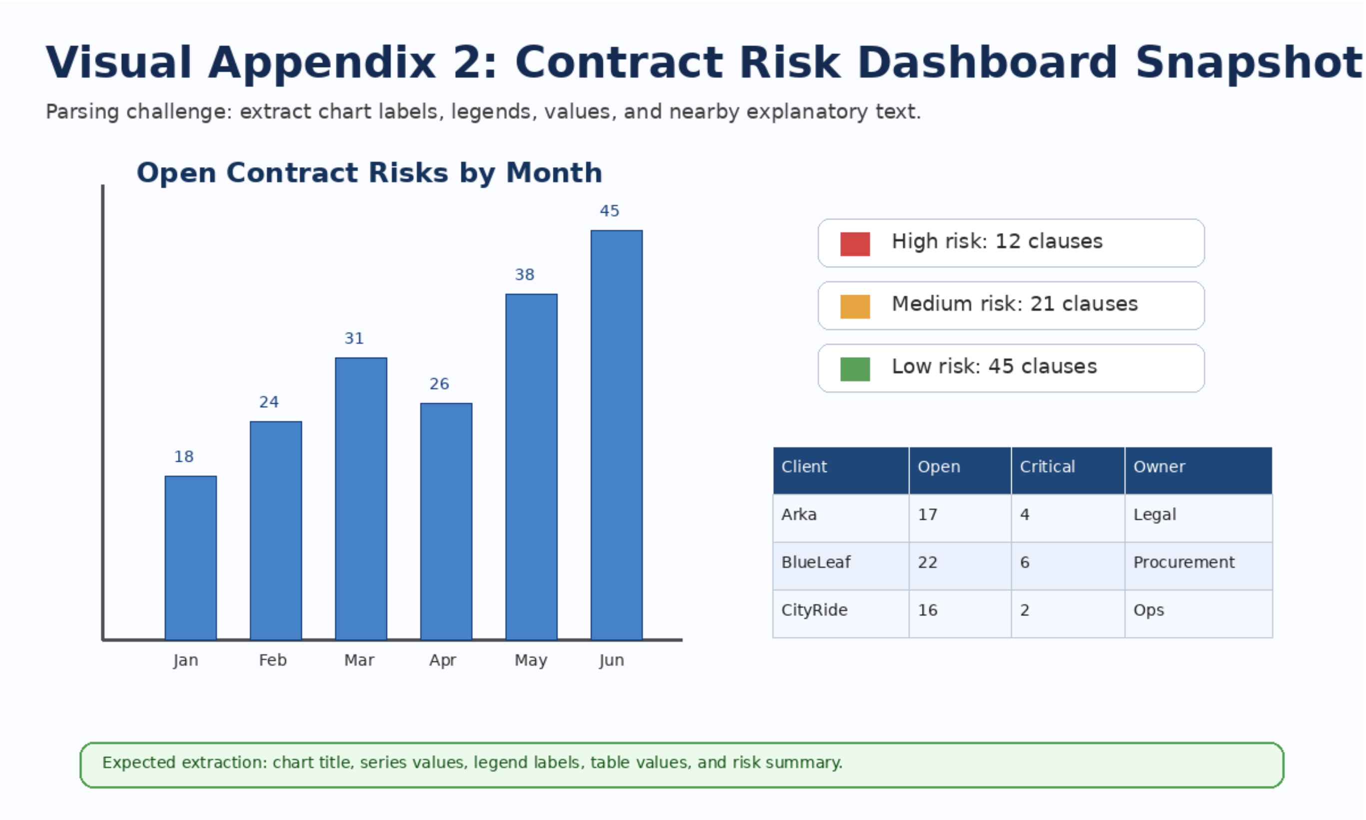


Added at the end for complex image parsing, OCR fallback, layout-aware extraction, and multimodal RAG testing.

Appendix E: Contract Risk Dashboard Image

Dashboard-style image containing a bar chart, legend, and small matrix. Useful for testing chart extraction, numeric value capture, and caption-aware indexing.

Parsing challenge: extract chart title, bar values, legend labels, and table values from an embedded image.

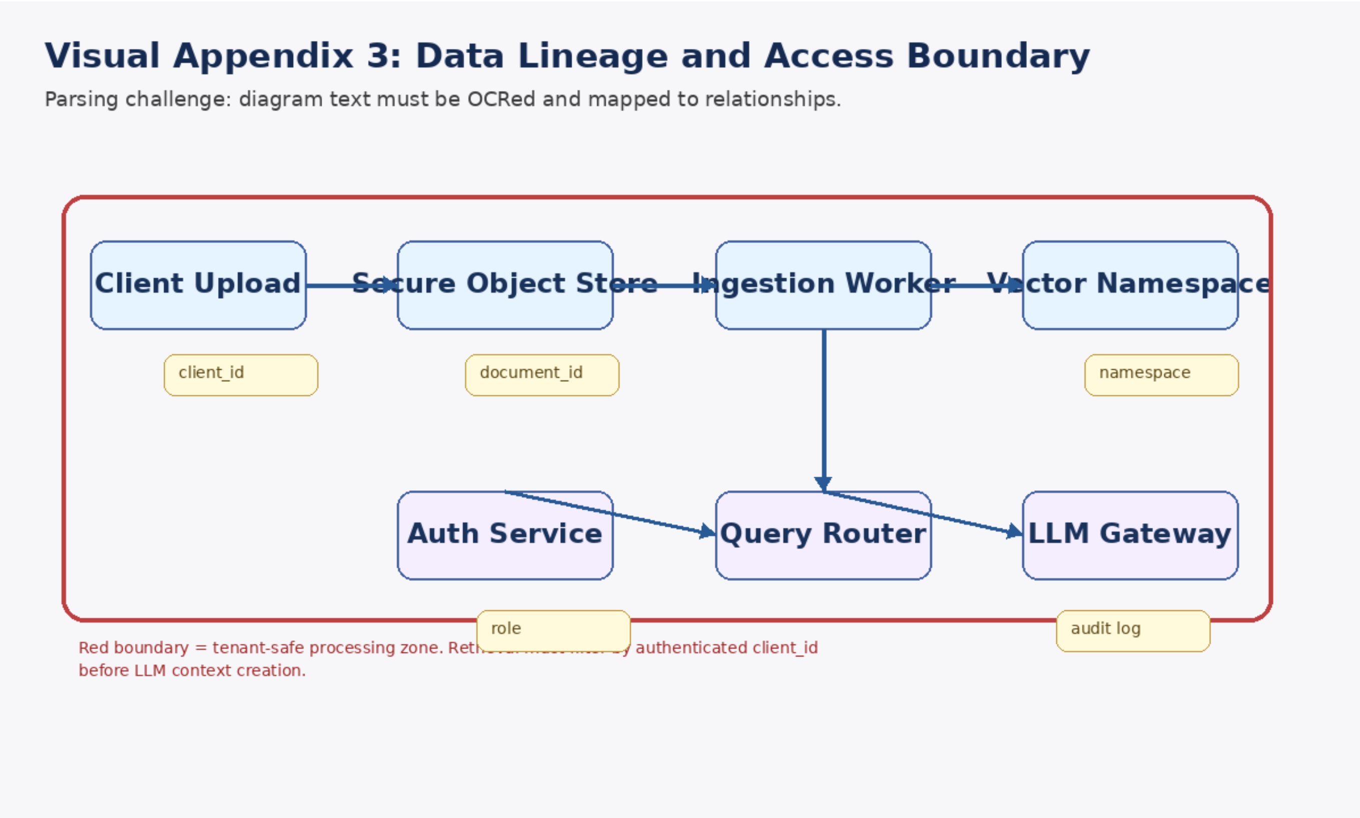


Added at the end for complex image parsing, OCR fallback, layout-aware extraction, and multimodal RAG testing.

Appendix F: Data Lineage and Access Boundary Image

Lineage-map style image with nodes, arrows, access boundaries, and metadata badges. Useful for diagram OCR and relationship extraction.

Parsing challenge: diagram text must be OCRed and mapped to relationships such as user auth, namespace, retrieval, and LLM gateway.



Added at the end for complex image parsing, OCR fallback, layout-aware extraction, and multimodal RAG testing.

Appendix G: Additional Scanned Invoice for OCR Testing

Synthetic scanned tax invoice with vendor details, customer details, invoice number, GSTIN, line items, tax split, total amount, payment terms, footer notes, approval stamp, and handwritten-style receipt text.

Parsing challenge: this page is intentionally embedded as an image-like scan. A normal text parser may miss invoice values unless OCR is enabled.

SCANNED TAX INVOICE

BlueLeaf Cloud Billing Services

GSTIN: 29AABCT2026P1Z8

No: INV-BLR-2026-0718

Date: 18 Jul 2026

Bill To:

School360 Learning Services Pvt Ltd
Tower B, Outer Ring Road
Bengaluru, Karnataka - 560103

Line Items

#	Description	Qty	Rate	Amount
1	Annual platform subscription	1	INR 95,000	95,000
2	Implementation and onboarding	1	INR 22,500	22,500
3	Support add-on / message pack	3,500	INR 0.40	1,400

Subtotal:	INR 118,900
CGST 9%:	INR 10,701
SGST 9%:	INR 10,701
Total Amount:	**INR 140,302**

Payment Terms:

Due within 15 days. Late fee may apply after due date.
Footer note: Amount includes taxes unless separately mentioned.
OCR challenge: faint stamp, rotated page, table lines, and handwritten approval.

APPROVED

Received by: K. Mehta

Added at the end for complex image parsing, OCR fallback, layout-aware extraction, and multimodal RAG testing.

Appendix H: Additional Scanned Utility Bill for OCR Testing

Synthetic scanned utility bill with meter-style charges, usage rows, tax values, total payable amount, payment terms, stamp, handwritten-style note, and noisy/rotated scan effects.

Parsing challenge: extract bill number, issuer, line items, usage quantity, tax split, total payable, and payment notes from a scanned image.

SCANNED UTILITY BILL

CityRide Depot Energy Board

GSTIN: 29AABCT2026P1Z8

No: BILL-CRM-2026-0881

Date: 18 Jul 2026

Bill To:

School360 Learning Services Pvt Ltd
Tower B, Outer Ring Road
Bengaluru, Karnataka - 560103

Line Items

#	Description	Qty	Rate	Amount
1	Electricity fixed charges	1	INR 1,250	1,250
2	Energy usage - peak units	842	INR 8.20	6,904
3	Energy usage - off-peak units	313	INR 5.70	1,784
4	Meter service adjustment	1	INR 340	340

Subtotal:	INR 10,278
CGST 9%:	INR 925
SGST 9%:	INR 925
Total Amount:	**INR 12,128**

Payment Terms:

Due within 15 days. Late fee may apply after due date.
Footer note: Amount includes taxes unless separately mentioned.
OCR challenge: faint stamp, rotated page, table lines, and handwritten approval.

APPROVED

Received by: K. Mehta

Added at the end for complex image parsing, OCR fallback, layout-aware extraction, and multimodal RAG testing.

Appendix I: Profile Image for Multimodal Parsing

This page adds a portrait-style image to test how a parser or multimodal RAG system handles photographic content, captions, image metadata, and surrounding text. Text-only loaders may extract the caption but cannot understand the visual content unless OCR, vision, or multimodal parsing is used.

Caption: Portrait-style instructor image with a bright background. Useful for testing image extraction, captioning, person detection, layout preservation, and multimodal document understanding.

Element	Parsing Challenge	Expected Handling
Portrait image	Pixels are not selectable PDF text.	Extract image object or create visual summary with a multimodal model.
Caption text	Caption should stay attached to image context.	Preserve caption near the image in reading order.
Image metadata	File name, page, and bounding box may be needed.	Store metadata for citation and retrieval.
RAG usage	Text-only chunks may miss visual details.	Use OCR or vision captioning before chunking if visual content matters.

Added at the end for complex image parsing, multimodal RAG testing, caption grounding, and visual-content extraction checks.